

Pranav Sharma

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Summary

Data-driven Computer Science student with hands-on experience designing, building, and deploying machine learning models and data pipelines for real-world business applications. Strong expertise in predictive modeling, time-series analysis, feature engineering, and statistical evaluation using Python, Pandas, NumPy, and Scikit-learn. Experience working with real-time and large-scale datasets, implementing end-to-end ML workflows including data ingestion, preprocessing, model training, validation, and deployment. Proven ability to translate raw data into actionable insights through data visualization and dashboards. Solid foundation in algorithms, data structures, and backend systems, with exposure to production-grade ML architectures.

Skills

Python, SQL, Pandas, NumPy, Scikit-learn, LightGBM, XGBoost, Regression, Classification, Clustering, Time Series Analysis, Feature Engineering, Data Cleaning, Model Evaluation, RMSE, R², Data Pipelines, ETL, Data Processing, FastAPI, PostgreSQL, Redis, Docker, Streamlit, Power BI, Matplotlib, Data Structures & Algorithms (DSA), REST APIs, Git, Agile/Scrum, Excel

Work Experience

Data Science Intern | May 2025 – August 2025

Jubilant FoodWorks Limited

- Built and automated an end-to-end multi-model sales forecasting pipeline in Python (Pandas, NumPy, Scikit-learn), reducing manual analysis time by 30% for faster decision making.
- Developed and evaluated regression and clustering models to predict Average Weekly Orders (AWOs), assessing 100+ prospective store locations to support data-driven expansion decisions.
- Performed feature engineering and model validation (train/test split, RMSE, R²) to select optimal models for deployment.
- Created an interactive Streamlit dashboard with geospatial heatmaps to visualize store performance potential for business stakeholders.

Machine Learning Project | September 2025 – February 2026

Real-Time S&P 500 Intraday Low Prediction System

- Built real-time ML system to predict probability that the S&P 500 intraday low has occurred using minute-level market data.
- Developed a scalable data pipeline for ingestion, feature engineering, and processing of financial time-series data.
- Trained and evaluated LightGBM/XGBoost models using walk-forward validation and probability calibration.
- Implemented a FastAPI backend with PostgreSQL and Redis for low-latency caching to get real-time prediction serving.
- Containerized the application using Docker and implemented modular architecture for real-time data processing.

Full Stack Development | January 2025 – April 2025

Personal E-commerce Project

- Developed a full-stack e-commerce application using Java, Spring Boot, and MySQL with integrated data handling and backend services.
- Designed and implemented RESTful APIs for product, order, and user management with validation and error handling.
- Modeled and optimized relational database schema with normalization, indexing, and foreign keys for efficient queries.
- Integrated frontend and backend systems, enabling dynamic data flow and real-time updates using JavaScript and API calls.

Education

Robert Morris University, Moon Township, PA

- **Bachelor of Science in Computer Science; Expected graduation May 2026**
- **GPA: 3.45; Dean's List: Fall 2024.**